

The empirical recurrence for OEIS A303893, recovered from its tail

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A303893 records “Empirical recurrence of order 92” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 154$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 99$. Its last coefficient is $c_{92} = 1152 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A303893 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 1, 2, 3 or 4 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 313, 842, 7421, 29479, 200144, 992789, 5930552, ...

The entry states:

Empirical recurrence of order 92 (see link above)

As of the “Last modified” line on the live entry (May 02 2018 08:03:35, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 154$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 154$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 92: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 92.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 308$ exact terms from $n = 7$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{92} c_i a(n-i),$$

c_1	1	c_{24}	501397545	c_{47}	−21065783830	c_{70}	527900413
c_2	48	c_{25}	565642776	c_{48}	−23922038716	c_{71}	34085228
c_3	13	c_{26}	−1000135267	c_{49}	31857066236	c_{72}	−9314390
c_4	−907	c_{27}	−359011958	c_{50}	−3608701502	c_{73}	64346445
c_5	−754	c_{28}	1219024511	c_{51}	−2172642411	c_{74}	−331324282
c_6	8758	c_{29}	922760591	c_{52}	3838600263	c_{75}	292288668
c_7	9064	c_{30}	−3093839411	c_{53}	−6476133216	c_{76}	−25664388
c_8	−50751	c_{31}	−1763459499	c_{54}	−17394927271	c_{77}	−128025544
c_9	−52242	c_{32}	8553423142	c_{55}	30085539703	c_{78}	61087812
c_{10}	199928	c_{33}	−3038009931	c_{56}	−2838841201	c_{79}	29360422
c_{11}	151847	c_{34}	−9901406014	c_{57}	−28074853434	c_{80}	−32103577
c_{12}	−648727	c_{35}	13725704978	c_{58}	26650414142	c_{81}	5536909
c_{13}	−81851	c_{36}	−1981471260	c_{59}	−7786192297	c_{82}	3269378
c_{14}	2344115	c_{37}	−11697598817	c_{60}	−6078694551	c_{83}	−436368
c_{15}	−1370780	c_{38}	8913471766	c_{61}	12694722793	c_{84}	−648838
c_{16}	−9850428	c_{39}	−8008534907	c_{62}	−11148251592	c_{85}	67679
c_{17}	10356344	c_{40}	13003918301	c_{63}	1340545217	c_{86}	36872
c_{18}	33559418	c_{41}	849898055	c_{64}	9588486126	c_{87}	−11152
c_{19}	−56842527	c_{42}	−25881727297	c_{65}	−11720377938	c_{88}	8696
c_{20}	−57834242	c_{43}	45744877781	c_{66}	3979311034	c_{89}	6732
c_{21}	215146792	c_{44}	−19475299518	c_{67}	2730506593	c_{90}	−8336
c_{22}	−56889766	c_{45}	−42700303794	c_{68}	−2328192764	c_{91}	1904
c_{23}	−486253920	c_{46}	56822375580	c_{69}	−153619604	c_{92}	1152

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 99$, and it is the recurrence OEIS A303893 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1x^{92-1} - \dots - c_{92}$. Its constant term is $-c_{92} = -1152$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 92, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1152.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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